

# Japan Math Olympiad 1991 — P3/5

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Let  $A$  be a positive 16-digit integer. Prove that you can select some sequence of consecutive digits from  $A$  such that their product forms a perfect square. (A single digit that is a perfect square is also valid).

### Solution

Let  $A = \overline{d_1 d_2 \cdots d_{16}}$ . Since  $0 \leq d_i \leq 9$  the product of some set of digits can only have prime factors 2, 3, 5 or 7. For prime  $p$ , and integer  $1 \leq n \leq 16$ , define

$$f_p(n) = v_p(d_1 \cdot d_2 \cdots d_n) \bmod 2.$$

Note that if there exists  $n$  such that  $f_2(n) = f_3(n) = f_5(n) = f_7(n) = 0$  then we are done because  $d_1 \cdot d_2 \cdots d_n$  is square since all the exponents are even in its prime factorisation. Suppose that such  $n$  doesn't exist. Consider the 4-tuple

$$T_k := (f_2(k), f_3(k), f_5(k), f_7(k)), \quad (k = 1, 2, \dots, 16).$$

Each  $f_p(k)$  is either 0, 1, and so there are  $16 - 1 = 15$  possible options for the tuple  $T_k$ , since we are excluding  $(0, 0, 0, 0)$ . Since we have  $16 > 15$  tuples  $T_1, T_2, \dots, T_{16}$ , by the Pigeonhole principle there must be two tuples  $T_i$  and  $T_j$ , ( $i < j$ ) such that  $T_i = T_j$ . Now because

$$v_p(d_i \cdot d_{i+1} \cdots d_j) = v_p(d_1 \cdot d_2 \cdots d_j) - v_p(d_1 \cdot d_2 \cdots d_{i-1})$$

and the fact that for  $p = 2, 3, 5, 7$  the two quantities on the RHS are the same parity, we deduce that  $d_i \cdot d_{i+1} \cdots d_j$  is square, since it only has even exponents in its prime factorisation. ■